

Peer Review File

Expanding drug targets for 112 chronic diseases using a machine learning-assisted genetic priority score



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REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

The paper explores how machine learning and genetic data can be utilized to identify new drug targets for chronic diseases. The study primarily leverages the data from the UK Biobank, using an approach called Machine Learning-assisted Genetic Priority Score (ML-GPS). This approach applies LightGBM models to predict the diagnosis of 112 chronic disease phecodes using features such as age, gender, lab results, lifestyle factors, and medical history. It also uses 239 features to predict the potential of gene-phecode pairs as drug targets, evaluating both current and novel drug indications.

There are a number of questions specifically regarding the methodologies used in the paper:

1. Have the authors tried some feature selection methods to select relevant features and remove redundancy to improve the predictive performance?
2. The choice of LightGBM is not well justified. How about other alternative machine learning algorithms except LightGBM? Have the authors conducted any preliminary experiments to benchmark and compare the performance of LightGBM with any alternative machine learning algorithms?
3. In the model section, the manuscript mentions that “We constructed machine learning models using LightGBM (version 4.0.0). We trained models to minimize log loss when predicting phecode diagnosis (encoded as zero or one). We selected the following parameters: {‘boosting_type’: ‘goss’, ‘num_iterations’: 1000, ‘learning_rate’: 0.01, ‘num_leaves’: 80, ‘min_data_in_leaf’: 100}.” Why did you set the parameters like this? Have you tried other parameter settings? Or have you applied any hyperparameter searching methods?
4. There are also other methods for addressing classification tasks such as the one in this study. Why did the authors only chose LightGBM and ElasticNet? More convincing and supportive data should be provided.

Reviewer #2 (Remarks to the Author):

Robert Chen et al. constructed ML-GPS with up to 13 integrated existing genetic evidence and continuous features, and trained the model to predict drug indications derived from Open Targets and externally testing it in SIDER. The study is well-designed and provides important insights into the landscape of potential drug targets for chronic diseases. Here are some comments that still need to be taken into consideration:

1. It would be better for the author to make the objective of applying ML-GPS more clear in the Introduction section. “However, recent advances have used machine learning to generate continuous metrics reflecting the full spectrum of disease liabilities, thereby enhancing the discovery of genetic variants associated with cardiovascular, pulmonary, and psychiatric diseases.”, “We use ML-GPS to generate predictions for 26,035 distinct genes and 112 phecodes for a total of 2,362,636 gene-phecode pairs and corroborate novel drug targets identified by ML-GPS using both known target-disease associations and manual screening.” After reading the full paper, I found that the gene-phecode pairs were obtained through mapping, genetic-disease associations were identified through genetic analyses. Despite machine learning models have been employed twice, the former one was used to select candidate phenotypes for the following analyses, and the later one was used to predict whether each distinct gene-phecode pair had an indicated drug (primary purpose of the study).
2. It may cause confusion that the author removed acute phecodes but kept cardiovascular ones. And it’s hard to define whether an acute condition can reflect chronic disease processes or not. Could the author make this clear?
3. For the 72 laboratory and vital measurements, can the author provide their missing rates across all participants so that the reader can have an idea about the imputation proportion?
4. The author has described that only 183,021 participants with GP records were included in the trained machine learning models. I was wondering whether the author used the same sub-population to run genetic analyses with regenie. If yes, have the author considered also including inpatient data to increase statistical power due to more cases included?
5. I was also wondering why the author run genetic analyses within UK Biobank participants instead of directly obtaining genetic associations from previous studies conducted in large-scale or diverse populations? Is that why the author also included existing genetic evidence in the models?
6. It was not very clear about how the non-directional genetic priority score was calculated. “We generated non-directional genetic priority scores by constructing ElasticNet logistic regression or LightGBM binary classification models with up to 13 features.” Can the author describe in more details regarding whether there exist any weights for the features when generating the genetic priority score?

7. For a given phenotype, if multiple variants were identified within the same gene, have the 13 features for the gene-disease pair incorporated the effects from all associated variants or only the effect of the most significant one?

Authors' response to all Reviewers:

We thank the reviewers for their thoughtful and constructive feedback on our manuscript. We have carefully addressed all comments and suggestions, which have helped us improve the clarity and rigor of our work. Major changes include additional analyses comparing different machine learning algorithms, clarification of our methodology and rationale for various choices, and expanded discussion of limitations. We believe these revisions have substantially strengthened the manuscript and hope the reviewers will find the revised version satisfactory.

Please note that we have reordered the manuscript such that the Methods section now follows the Discussion section, in accordance with *Nature Communications* guidelines.

Reviewer #1 (Remarks to the Author):

The paper explores how machine learning and genetic data can be utilized to identify new drug targets for chronic diseases. The study primarily leverages the data from the UK Biobank, using an approach called Machine Learning-assisted Genetic Priority Score (ML-GPS). This approach applies LightGBM models to predict the diagnosis of 112 chronic disease phecodes using features such as age, gender, lab results, lifestyle factors, and medical history. It also uses 239 features to predict the potential of gene-phecode pairs as drug targets, evaluating both current and novel drug indications.

There are a number of questions specifically regarding the methodologies used in the paper:

1.1. Have the authors tried some feature selection methods to select relevant features and remove redundancy to improve the predictive performance?

Thank you for this suggestion. We did not perform feature selection before model training (i.e. pre-training feature selection) because LightGBM already performs internal feature selection by selecting the best splits during tree construction. Gradient boosting models are also immune to multicollinearity as only one of multiple redundant features will be selected by the model when deciding upon a split. We acknowledge that feature selection is useful for reducing model complexity, but we do not believe this is relevant here as we apply our models only in the UK Biobank.

As a sensitivity analysis, we trained models using only the top 1/3 or top 2/3 of features as determined by the improvement in log loss attributable to each feature, and included the results in **Table S3**. For all phecodes, there was either a decrease or no substantial change in all performance metrics, confirming that pre-training feature selection is not necessary and that the LightGBM internal feature selection already optimizes this step .

We added the following sentence to the methods clarifying this: "We did not perform pre-training feature selection because LightGBM performs internal feature selection during tree construction;

supporting this, models including only important features did not have substantially different performance from models with all features (**Table S3**)."

1.2. The choice of LightGBM is not well justified. How about other alternative machine learning algorithms except LightGBM? Have the authors conducted any preliminary experiments to benchmark and compare the performance of LightGBM with any alternative machine learning algorithms?

We selected gradient boosting models because prior studies in the machine learning literature have demonstrated that gradient boosting models consistently outperform all other models on tabular datasets like ours, including in a 2022 benchmarking paper that included random forest and recent deep tabular learning models.¹ Among gradient boosting models, XGBoost and LightGBM are the two major packages available; we selected LightGBM as it is substantially faster and more memory efficient than XGBoost (an important consideration as we were training hundreds of models) while having equal or better accuracy.^{2,3}

As a sensitivity analysis, we also included comparisons between LightGBM, XGBoost, and random forest models for both the phecode diagnosis prediction models and ML-GPS. In all cases, LightGBM either significantly outperformed or was comparable to XGBoost and random forest while being 2-4x faster to train.

We have made the following changes to the manuscript:

1. Regarding phecode diagnosis prediction models: "LightGBM models were robust to feature selection (**Table S3**), and either outperformed or were comparable to XGBoost and random forest models (**Table S4**)."
2. Regarding ML-GPS (prediction of drug indications): "As a sensitivity analysis, we also compared the LightGBM-based GB (CE phase weights) model with XGBoost and random forest models; LightGBM outperformed the latter two models in AUPRC in both Open Targets and SIDER (**Figure S3; Table S13**)."

1.3. In the model section, the manuscript mentions that "We constructed machine learning models using LightGBM (version 4.0.0). We trained models to minimize log loss when predicting phecode diagnosis (encoded as zero or one). We selected the following parameters: {'boosting_type': 'goss', 'num_iterations': 1000, 'learning_rate': 0.01, 'num_leaves': 80, 'min_data_in_leaf': 100}." Why did you set the parameters like this? Have you tried other parameter settings? Or have you applied any hyperparameter searching methods?

Yes, we used GridSearchCV from scikit-learn to tune the 'boosting_type' and 'num_leaves' parameters, which are the two major parameters that affect model performance.

- boosting_type controls the algorithm used by LightGBM to construct trees. For all values of num_leaves, goss outperformed all other algorithms.
- num_leaves controls model complexity. Higher values generally yield better accuracy at the cost of increased training time, and values < 100 are recommended to prevent

overfitting. Across all phecodes we also did not observe increased accuracy for values above 80, so we selected 80 as a uniform, conservative value.

The remaining parameters have minor effects on model performance and we instead followed recommendations provided by LightGBM.⁴

- `min_data_in_leaf` prevents overfitting by controlling the minimum amount of data points that must be present in a leaf node in each tree; we set this to 5x the default value (20) due to the large size of our dataset.
- Using a small `learning_rate` and large `num_iterations` is also recommended for better accuracy; we decreased `learning_rate` from 0.1 (default) to 0.01 and increased `num_iterations` from 100 (default) to 1000. Please note that `num_iterations` is a maximum limit, and we enabled early stopping such that model training automatically stops if there is no improvement in log loss after 10 iterations. In all cases model training was completed in under 300 iterations so these parameters did not limit the accuracy of our model.

We clarified this in the methods as follows: “We tuned ‘`boosting_type`’ (options ‘`gbdt`,’ ‘`goss`,’ and ‘`dart`’) and ‘`num_leaves`’ (increments of 10 from 10 to 100) to optimize AUROC using the `GridSearchCV` function from `scikit-learn` and selected the other three parameters based on LightGBM recommendations.”

1.4. There are also other methods for addressing classification tasks such as the one in this study. Why did the authors only chose LightGBM and ElasticNet? More convincing and supportive data should be provided.

As mentioned in our response to comment 1.2, LightGBM remains state-of-the-art for tabular data, and we added comparisons between LightGBM, XGBoost, and random forest models in **Table S4**, **Table S13**, and **Figure S3**. The reason we included ElasticNet is because our original implementation of GPS used logistic regression, and we wanted to see whether changing from logistic regression to gradient boosting would improve performance (**Figure 3a-d**). We specifically used ElasticNet regularization because of multicollinearity of features here (especially between P, C, and B features).

Reviewer #2 (Remarks to the Author):

Robert Chen et al. constructed ML-GPS with up to 13 integrated existing genetic evidence and continuous features, and trained the model to predict drug indications derived from Open Targets and externally testing it in SIDER. The study is well-designed and provides important insights into the landscape of potential drug targets for chronic diseases. Here are some comments that still need to be taken into consideration:

2.1. It would be better for the author to make the objective of applying ML-GPS more clear in the Introduction section. “However, recent advances have used machine learning to generate continuous metrics reflecting the full spectrum of disease liabilities, thereby

enhancing the discovery of genetic variants associated with cardiovascular, pulmonary, and psychiatric diseases.”, “We use ML-GPS to generate predictions for 26,035 distinct genes and 112 phecodes for a total of 2,362,636 gene-phecode pairs and corroborate novel drug targets identified by ML-GPS using both known target-disease associations and manual screening.” After reading the full paper, I found that the gene-phecode pairs were obtained through mapping, genetic-disease associations were identified through genetic analyses. Despite machine learning models have been employed twice, the former one was used to select candidate phenotypes for the following analyses, and the later one was used to predict whether each distinct gene-phecode pair had an indicated drug (primary purpose of the study).

We agree with your suggestion and have revised the second paragraph to explicitly state the two stages of machine learning model application as follows: “Building upon our recent Genetic Priority Score (GPS) framework,⁵ which demonstrated the efficacy of using genetic features to prioritize drug targets, we introduce a novel machine learning-assisted version of the GPS (ML-GPS) applied to 112 chronic disease phecodes. Our approach employs machine learning in two key stages: first, to improve phenotyping via continuous metrics of disease liability, and second, to predict drug indications using genetic associations with these phenotypes as well as existing genetic evidence. In the second stage, ML-GPS uses complementary sources of evidence to assign each gene-phecode pair a probability of having an indicated drug, allowing researchers to select high-scoring genes for further screening.”

2.2. It may cause confusion that the author removed acute phecodes but kept cardiovascular ones. And it’s hard to define whether an acute condition can reflect chronic disease processes or not. Could the author make this clear?

Thank you for this recommendation. We have clarified that we included only five phecodes which, although are acute events, are widely recognized as reflecting chronic disease processes, and provided references on the pathophysiology of these processes. Specifically, myocardial infarction and (ischemic) stroke are included in all major definitions of atherosclerotic cardiovascular disease (ASCVD); arterial thrombosis is likewise caused by atherosclerosis; and arterial dissection is linked to vasa vasorum dysfunction and connective tissue disorders.

While we agree there is some subjectivity to this inclusion, we believe that our preliminary machine learning models also serve as a filter for whether a disease represents a chronic disease process or not. These models utilized only laboratory and vital measurements from a single timepoint for each participant (UK Biobank enrollment date) to predict both prevalent and incident diagnoses, which were often many years before or after the enrollment date, respectively. For acute diseases that do not represent chronic processes, we would not expect accurate predictions, as most cases would likely have “normal” measurements at enrollment, being several years prior to or following the acute event.

We clarified this as follows:

- **Inclusion of phecodes:** “However, we retained five acute phecodes that represent chronic disease processes: four of them [myocardial infarction (CV_404.1), cardiac arrest (CV_420), stroke (CV_431.1), arterial embolism and thrombosis (CV_438.4)] that reflect atherosclerosis, and one [arterial dissection (CV_438.4)] that reflects vasa vasorum dysfunction.⁶ This filtering yielded 386 phecodes.”
- **Preliminary models serving as a filter for chronic disease processes:** “As an initial filter for evaluating machine learning model performance, we constructed preliminary models using age, sex, and 72 laboratory and vital measurements for the 386 chronic disease phecodes identified earlier. We retained 112 phecodes for which the area under the receiver operating characteristic curve (AUROC) was ≥ 0.70 and the area under the precision-recall curve (AUPRC) exceeded the phecode's prevalence. This step was also intended to select phecodes associated with chronic physiological changes. Since many disease diagnoses occur several years before or after a participant's enrollment in the UK Biobank (when measurements are recorded), diseases not associated with chronic changes would likely not be accurately predicted by the model.”

We also added the following as a limitation in the discussion: “Second, this study does not comprehensively cover all chronic diseases. Our phecode selection process included a semi-subjective manual screening where valid phecodes may have been erroneously removed. Further, we excluded many chronic disease phecodes, especially those in the musculoskeletal and sense organ categories, because they could not be accurately predicted using available phenotypic data. The latter issue also emphasizes the importance of complete phenotypic data, with the majority of UK Biobank lacking imaging, audiometric, and ophthalmic data.”

2.3. For the 72 laboratory and vital measurements, can the author provide their missing rates across all participants so that the reader can have an idea about the imputation proportion?

Thank you for this suggestion. We have added the missingness rates in **Table S20** and discussed them as follows in the methods: “44 features had missingness rates below 1% and all had missingness rates below 10% except for direct bilirubin (15.1%) and lipoprotein A (20.8%) (**Table S20**).”

2.4. The author has described that only 183,021 participants with GP records were included in the trained machine learning models. I was wondering whether the author used the same sub-population to run genetic analyses with regenie. If yes, have the author considered also including inpatient data to increase statistical power due to more cases included?

We apologize for any confusion. To clarify, we trained machine learning models to predict phecode diagnoses only among 183,021 participants with GP records, but used them to generate predictions for all 396,605 participants. For genetic analyses of observed case/control status, we indeed used both GP records and inpatient diagnoses to maximize statistical power, and we performed this analyses on all participants, not only the subset with GP records.

To clarify this, we have made several changes to the methods:

- Regarding machine learning models to predict phecode diagnoses: “Thus, we trained models only on the 183,021 participants with GP records, using both GP and inpatient records for these participants to assign case/control status. We then used trained models to generate predictions for the remaining 213,584 participants without GP records. For consistency between those with and without GP records, we only used inpatient diagnoses, which were available for all participants, as features for the machine learning model.”
- Regarding genetic analyses: “We performed each analysis with three different phenotypes among all 396,605 participants: observed phecode case/control status (“P”), binarized model probabilities (“B”), and continuous model probabilities (“C”). We defined observed case/control status using both GP records and inpatient diagnoses.”

2.5. I was also wondering why the author run genetic analyses within UK Biobank participants instead of directly obtaining genetic associations from previous studies conducted in large-scale or diverse populations? Is that why the author also included existing genetic evidence in the models?

We ran our own genetic association analyses in the UK Biobank because one of the main goals of this paper was to compare genetic analyses with observed case/control status (“P”) with machine learning-predicted phenotypes (“C” and “B”), and to do so rigorously, we needed to perform these genetic association analyses using the same methods and among the same participants.

Additionally, as you mentioned, the Locus-to-gene (L2G) feature in our models developed by OpenTargets already incorporates external genetic evidence from other large-scale, diverse populations, including 300 studies with 14,013 lead SNPs from the GWAS Catalog.⁷ Thus, we felt it was not necessary to include non-UK Biobank genetic association results from the GWAS Catalog as additional features.

2.6. It was not very clear about how the non-directional genetic priority score was calculated. “We generated non-directional genetic priority scores by constructing ElasticNet logistic regression or LightGBM binary classification models with up to 13 features.” Can the author describe in more details regarding whether there exist any weights for the features when generating the genetic priority score?

We revised the first paragraph of the “Genetic priority scores” section of the methods to include a broader overview of the methodology and specify that feature weights were not pre-determined: “We constructed ElasticNet logistic regression and LightGBM binary classification models to predict whether each gene-phecode pair has an indicated drug and used continuous prediction probabilities from these models as genetic priority scores. These scores are non-directional as they do not predict whether a drug with an activator or inhibitor mechanism is required. We trained models using the larger Open Targets dataset (112,274 pairs) and

externally tested them in the smaller SIDER dataset (58,674 pairs). Models included up to 13 features: four of these features represented existing genetic evidence (EVA-ClinVar, HGMD, OMIM, L2G). The other nine features represented genes identified from the common, rare, and ultra-rare variant analyses for P, B, and C phenotypes. Feature weights and importances were not pre-defined and were determined automatically by each model.”

Additionally, for the ElasticNet model, we have now provided regression coefficients in **Table S25** and added the following sentence to the methods: “ElasticNet regression coefficients are available in **Table S25**.” LightGBM does not have “weights” or coefficients, but feature importances as determined by SHAP analysis are available in **Figure S4** and discussed in the “Construction of ML-GPS” section of the results.

2.7. For a given phenotype, if multiple variants were identified within the same gene, have the 13 features for the gene-disease pair incorporated the effects from all associated variants or only the effect of the most significant one?

We thank the reviewer for this important point. If multiple variants were identified within the same gene, we only considered the most significant variant; we state this in the methods as follows: “We compared this with continuous encoding as follows: for EVA-ClinVar, HGMD, and OMIM, we assigned each gene the number of distinct variants for that gene; for L2G, we assigned each gene the highest L2G score of all variants for that gene; and for the P, B, and C features, we assigned each gene the highest $-\log_{10}(\text{p-value})$ from the genetic association results of all variants or tests for the gene.”

We did not consider the effects of all associated variants for common and rare variant phenotypes due to the difficulty of representing this information as a single number. The ultra-rare variant associations, however, were derived from gene burden testing, and represent the aggregate effects of all deleterious missense and protein truncating variants in each gene.

References

1. Grinsztajn, L., Oyallon, E. & Varoquaux, G. Why do tree-based models still outperform deep learning on typical tabular data?
2. Experiments — LightGBM 4.4.0.99 documentation.
<https://lightgbm.readthedocs.io/en/latest/Experiments.html>.
3. Ke, G. *et al.* LightGBM: A Highly Efficient Gradient Boosting Decision Tree. in *Advances in Neural Information Processing Systems* vol. 30 (Curran Associates, Inc., 2017).
4. LightGBM Parameters Tuning. <https://lightgbm.readthedocs.io/en/latest/Parameters-Tuning.html>.
5. Duffy, Á. *et al.* Development of a human genetics-guided priority score for 19,365 genes and 399 drug indications. *Nat Genet* **56**, 51–59 (2024).
6. Bax, M. *et al.* Arterial dissections: Common features and new perspectives. *Front Cardiovasc Med* **9**, 1055862 (2022).
7. Mountjoy, E. *et al.* An open approach to systematically prioritize causal variants and genes at all published human GWAS trait-associated loci. *Nat Genet* **53**, 1527–1533 (2021).

REVIEWERS' COMMENTS

Reviewer #1 (Remarks to the Author):

In this revised manuscript, the authors have adequately addressed the reviewers' points and accordingly improved the quality of their work. I am more than happy to recommend this revised version for publication in Nature Communications.

Reviewer #2 (Remarks to the Author):

I am fully satisfied of the explanations and details provided and have no further comment.

Reviewer #2 (Remarks on code availability):

The code seems valid and helpful.